AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 03

Prompt Engineering

Basics

Lesson 03 of 30 | AI & Productivity Program

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- The 5-component anatomy of a powerful AI prompt
- Zero-shot, few-shot and chain-of-thought techniques
- Why your prompts fail — and how to fix them in seconds
- 15 copy-paste prompt templates for work, study and engineering
- How temperature and AI parameters shape every response

01 | What Is Prompt Engineering?

A prompt is any instruction you give an AI model. Prompt engineering is the skill of crafting those instructions to get **accurate, useful, and consistent outputs**. The same AI model given a weak prompt produces mediocre results. Given a well-crafted prompt, it produces expert-level work. The model did not change — only the input did.

This is the single most practical AI skill you can learn in 2026. It applies to ChatGPT, Claude, Gemini, and every other LLM. Mastering it does not require technical knowledge — it requires understanding **how language models think**.

> **Why Prompt Engineering Matters**
>
> A Stanford study found that the same GPT-4 model scored 48% on a legal exam with a basic prompt — and 75% with an optimised prompt. Same model, same exam. The prompt was the only variable. That 27-point gap is the value of this skill.

02 | The 5-Component Anatomy of a Powerful Prompt

Every great prompt shares the same underlying structure. You do not need all five components every time — but knowing each one and when to use it is what separates average AI users from people who get consistently excellent results.

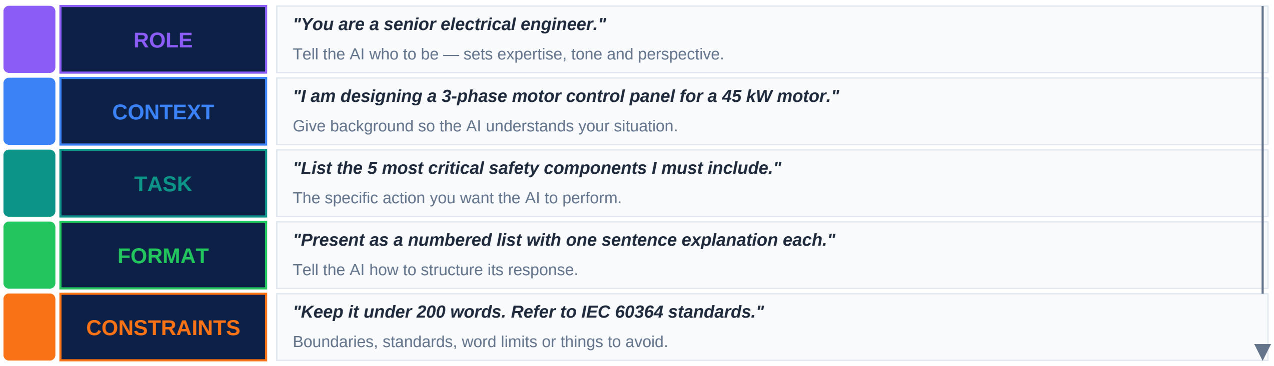

Figure 1 — The 5-Component Anatomy of a Powerful Prompt

Component	Question It Answers	Without It → Problem
Role	Who should the AI be?	AI gives a generic answer instead of expert-level depth
Context	What is the situation and background?	AI makes assumptions that do not match your actual problem
Task	What exactly do I want the AI to do?	AI interprets the task differently than you intended
Format	How should the answer be structured?	AI gives a wall of text when you needed a table or bullet list
Constraints	What limits, standards or rules should apply?	AI ignores word limits, uses wrong standards, goes off-topic

03 | Core Prompting Techniques

Beyond the basic structure, several specific techniques dramatically improve output quality for different task types.

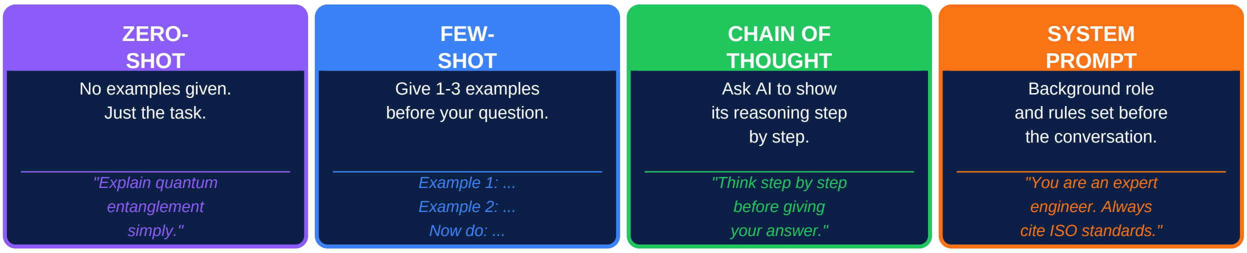

Figure 2 — Four Core Prompting Techniques

Technique	Best For	Example Trigger Phrase
Zero-shot	Quick questions, definitions, summaries, creative tasks	"Explain X" / "Write a Y" / "List the top Z"
Few-shot	Classification, formatting tasks, style matching, translation	"Here are 3 examples of the format I want: [examples]. Now do this: [task]"
Chain-of-thought	Maths, logic, multi-step reasoning, complex decisions	"Think through this step by step before giving your answer."
System prompt	Consistent role across a long conversation, custom AI assistant setup	"You are [role]. Your rules are [rules]. Always [behaviour]."
Role + Audience	Adjusting complexity for a specific reader	"Explain [topic] to a [audience] who has never studied [subject]."
Iterative refinement	Any task — first draft then improve	"Now make it more concise." / "Add a real-world example to point 3."

04 | Weak Prompt vs Strong Prompt

The fastest way to understand prompt engineering is to see the same question asked badly and then well. The diagram below shows a real engineering example.

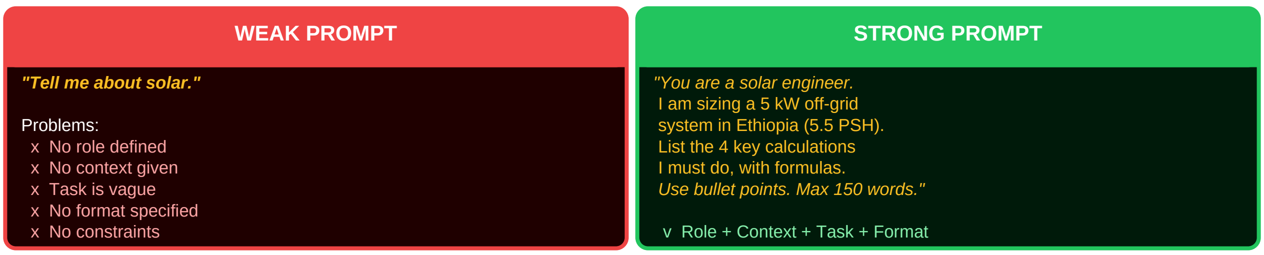

Figure 3 — Weak vs Strong Prompt: Same Topic, Radically Different Results

The Golden Rule of Prompt Writing

> Treat AI like a brilliant new employee on their first day. They are smart and capable — but they know nothing about your specific situation. The more precisely you brief them, the better they perform. Vague instructions produce vague results. Every time.

05 | 15 Prompt Templates You Can Use Today

Use Case	Copy-Paste Template
Explain a concept	"You are an expert in [field]. Explain [concept] to a [beginner/intermediate] in under 150 words using a real-world analogy."
Summarise a document	"Summarise the following in 5 bullet points, focusing on the most actionable insights: [paste text]"
Engineering calculation	"You are a [electrical/mechanical/HVAC] engineer. Walk me step by step through calculating [quantity] for [scenario]. Show all formulas."
Compare two options	"Compare [Option A] vs [Option B] for [use case]. Use a table with columns: Feature | Option A | Option B | Winner."
Write a professional email	"Write a professional email to [recipient] requesting [action]. Tone: [formal/friendly]. Length: under 120 words."
Debug / fix code or logic	"Find the error in this [code/calculation/argument] and explain why it is wrong, then give the corrected version: [paste content]"
Study plan	"Create a 30-day study plan to learn [topic] starting from [level]. Include daily topics and 3 resources per week."
Generate questions	"Generate 10 exam-style questions on [topic] at [beginner/intermediate/advanced] level with model answers."
Devil's advocate	"I believe [position]. Give me the 5 strongest counterarguments to this view, even if you agree with me."
Translate and simplify	"Translate the following technical document into plain English a non-engineer can understand: [paste text]"
Risk analysis	"You are a risk engineer. List the top 10 risks for [project/system], rate each 1-5 for likelihood and impact, and suggest mitigations."
First-principles breakdown	"Break [concept/product/problem] down to its most fundamental components using first-principles thinking."
Write a report section	"Write a [Introduction/Methodology/Conclusion] section for a report on [topic]. Formal academic tone. 200-250 words."
Critique my work	"Critique the following [essay/design/plan/code] as a strict expert. Be specific. Do not just praise — identify every weakness: [paste]"
Roleplay scenario	"Roleplay as [expert/interviewer/client]. I will [describe my role]. Challenge my thinking on [topic]. Start with a hard question."

06 | Understanding Temperature and AI Parameters

Parameter	Range	Low Value → Effect	High Value → Effect	Best Setting For
Temperature	0.0 – 2.0	Deterministic, repetitive, safe. Same input → same output.	Creative, surprising, random. More variation in outputs.	Facts/maths/code: 0.0–0.3. Creative writing: 0.7–1.2
Max tokens	1 – context limit	Short response — may cut off mid-sentence.	Long detailed response — may include padding.	Match to your actual needed length. Short = 200, detailed = 1 000+

Top-P (nucleus)	0.0 – 1.0	Only considers most probable next tokens — predictable.	Considers a wider vocabulary — more varied language.	Leave at default (0.9–1.0) unless you have a specific need
System prompt	Text field	No role defined — AI is a generic assistant.	Strong, detailed role — AI behaves as a specialist.	Always set for professional or repeated use cases

What comes next — AI-004

Advanced Prompt Engineering: multi-turn conversations, prompt chaining, structured output (JSON/tables), working with long documents, and building your own custom GPT/Claude agent.